

Exploring the potential of retinal organoids for modeling the XLRS-associated *RS1* variants

Andrea Heredero Berzal^{1,2}, Anneloor L M A Ten Asbroek², Jacoline B. ten Brink², Céline Koster^{1,2,3}, Camiel J F Boon^{1,4}.

Author affiliations: 1 Department of Ophthalmology, Amsterdam UMC, University of Amsterdam (UvA), 1105 AZ Amsterdam, The Netherlands. 2 Department of Human Genetics, Amsterdam UMC, University of Amsterdam (UvA), 1105 AZ Amsterdam, The Netherlands. 3 Emma Center for Personalized Medicine, Amsterdam UMC, 1105 AZ Amsterdam, The Netherlands. 4 Department of Ophthalmology, Leiden University Medical Center (LUMC), 2333 ZA Leiden, The Netherlands.

Variants in the *RS1* gene are known to cause X-linked retinoschisis (XLRS), characterized by the splitting of the inner retinal layers. This schisis is a structural defect that leads to loss of retinal organization and eventually disrupted communication between photoreceptors and bipolar cells. Although animal models have proved valuable for studying disease pathomechanisms and have shown the potential of gene therapy for XLRS, no treatment is currently available for these patients. Human-relevant models to understand the underlying biology of XLRS may be necessary for that. In this context, retinal organoids have emerged as a valuable model for studying inherited retinal dystrophies. In this study, we characterize two pathogenic variants in the *RS1* gene: a missense variant c.214G>A (p.E72K) in exon 4 and a deletion in exon 3, c.78+365_c.184+607del (del.ex3). These variants were modeled in retinal organoids generated using a 3D-embedding system. Our findings demonstrate that patient-derived retinal organoids exhibited reduced RS1 protein presence compared to control organoids, while lamination remains unaffected. Furthermore, reduced RS1 protein levels appeared to be linked to functional dysregulation (as measured by microelectrode arrays), as well as disruptions of the photoreceptor developmental timeline and ultrastructure. Nevertheless, although the simplified architecture of organoids can be beneficial for modeling certain phenotypic characteristics, this may not be the case for schisis formation, since the variants modeled in this study did not result in the formation of this characteristic XLRS hallmark. In this study, we further assess the current limitations of retinal organoids and the critical components needed to advance *in vitro* XLRS modeling.